

Pingchuan Ma

UNIVERSITY OF MUNICH · COMPUTER VISION & LEARNING GROUP

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Research Interests

- Improving the training, inference, and practical deployment of generative models like diffusion and flow matching.
- Applying generative model priors beyond synthesis, such as depth estimation, 3D modeling, dynamics of artistic style evolution, and more.
- Studying alignment and structure across data modalities, and building representations that support controllable generation, editing, and downstream visual understanding.

Research Experience

Computer Vision & Learning Group, University of Munich

Research Assistant | MCML PhD Student

Nov. 2021 – present

Focus on diffusion and flow-based generative models, multimodal representation learning, and generative model priors beyond synthesis.

Bayer AG

Industrial Collaboration

Jul. 2023 – Jun. 2026

Adapt generative models for plant image synthesis across controlled and in-the-wild conditions. Develop multimodal representations for fungal microscopy data under treatment variation, with attention to interpretability and domain language.

CLINC 5.1

Researcher

Oct. 2021 – Jun. 2023

Worked on representation learning for urological tumour data across CT, MRI, and histopathology under limited-data conditions.

Computer Vision Lab, Heidelberg University

Research Assistant / Intern

Aug. 2018 – Sep. 2020

Worked on deep metric learning and content-aware style transfer, resulting in TPAMI 2021 and CVPR 2019 publications.

Selected Publications

- Johannes Schusterbauer, Ming Gui, Yusong Li, **Pingchuan Ma**, Felix Krause, Björn Ommer. “Denoising, Fast and Slow: Difficulty-Aware Adaptive Sampling for Image Generation”. CVPR 2026 [↗](#)
- Michael Fuest, **Pingchuan Ma**, Ming Gui, Johannes Schusterbauer, Vincent Tao Hu, Björn Ommer. “Diffusion Models and Representation Learning: A Survey”. TPAMI 2026 [↗](#)
- Pingchuan Ma**^{*}, Ming Gui^{*}, Johannes Schusterbauer, Xiaopei Yang, Olga Grebenkova, Vincent Tao Hu, Björn Ommer. “Stochastic Interpolants for Revealing Stylistic Flows across the History of Art”. ICCV 2025 [↗](#)
- Pingchuan Ma**^{*}, Xiaopei Yang^{*}, Ming Gui, Yusong Li, Felix Krause, Johannes Schusterbauer, Björn Ommer. “SCFlow: Implicitly Learning Style and Content Disentanglement with Flow Models”. ICCV 2025 [↗](#)
- Pingchuan Ma**^{*}, Lennart Rietdorf^{*}, Dmytro Kotovenko, Vincent Tao Hu, Björn Ommer. “Does VLM Classification Benefit from LLM Description Semantics?”. AAAI 2025 [↗](#) Invited talk at Imageomics-AAAI-25 [↗](#).
- Ming Gui, Johannes Schusterbauer, Ulrich Prestel, **Pingchuan Ma**, Dmytro Kotovenko, Olga Grebenkova, Stefan A. Baumann, Vincent Tao Hu, Björn Ommer. “DepthFM: Fast Monocular Depth Estimation with Flow Matching”. AAAI 2025 Oral [↗](#)
- Johannes Schusterbauer^{*}, Ming Gui^{*}, **Pingchuan Ma**^{*}, Nick Stracke, Stefan A. Baumann, Björn Ommer. “Boosting Latent Diffusion with Flow Matching”. ECCV 2024 Oral [↗](#)
- Artsiom Sanakoyeu, **Pingchuan Ma**, Vadim Tschernezki, Björn Ommer. “Improving Deep Metric Learning by Divide and Conquer”. TPAMI 2021 [↗](#)

Education

University of Munich / MCML

Munich, Germany

PhD Student in Computer Vision and Machine Learning

Nov. 2021 – expected Aug./Sep. 2026

Supervisor: Prof. Dr. Björn Ommer.

Heidelberg University

Heidelberg, Germany

M.Sc., Applied Computer Science

Sep. 2017 – Jan. 2021

Grade: 1.5, German scale.

Guangzhou University of Chinese Medicine

Guangzhou, China

B.Eng., Computer Science and Information Technology

Sep. 2011 – Jun. 2015

GPA: 3.35 / 4.0, top 15%.

Skills

Programming & ML

Python, PyTorch, Hugging Face Accelerate, DDP, Lightning, SLURM, Linux, AI-assisted coding.

Research Tooling

Large-scale ML experimentation, ablations, benchmarking, image/video generation, vision-language evaluation.

Languages

English fluent; German B2; Mandarin and Cantonese native.

Miscellaneous

Kayaking, Hiking, and all other nature-related sports.